Corvinus University of Budapest

Mortality forecasting with the Lee-Carter Model using two different estimation methods

Authors: Bendegúz Birkmayer, Bojta Rácz,

Kristóf Légrádi, Roland Tuboly

Social Data Science MSc

Instructor: László Kovács

Computational Statistics

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1. Summary for UN demographers

Using Human Mortality Database data (HMD, 2025) on deaths, exposures, and mortality rates for males and females in 30 countries, 1960–2019, we addressed the question of which estimation method (SVD or Poisson MLE) of the Lee-Carter model gives lower out-of-sample forecast error.

We forecast mortality rates for 2005-2019 (pre-COVID-19) using models fitted to observed data between 1960 and 2004. We then derived the Root Mean Square Error (RMSE) aggregated over ages and years within each country-sex pair from the differences between observed and forecast mortality rates. The results show that in our out of sample test (2005–2019) across 30 countries, SVD had lower RMSE in most country-sex groups. We also compared performance within selected country groups (Mediterranean and Post-Socialist) to address the issue of different underlying mortality mechanisms and patterns, but SVD still produced better results. We obtained a better result for MLE only for Denmark. The gap may reflect higher variance at older ages due to smaller exposures and shocks.

The main implication of our findings on mortality forecasting and pension schemes is that given our evaluation window and metric, SVD is a safer default for these countries, but results can depend on optimizer settings, weighting, and the forecast horizon. Precise predictions that do not over- or underestimate rates are necessary to determine how many pensioners a given country must support each year. Furthermore, it is also crucial to predict how long an individual will receive pension, how many individuals reach retirement age and how likely it is that in the next year, pension payments cease upon death. Different regions have different mortality patterns (e.g. in post-socialist countries many men do not even reach retirement age), but our results show that SVD performed better for most countries. Denmark, where MLE performed better in our implementation, was the only exception.

Long horizon projections can also be affected by rotation, the tendency for mortality decline to slow at younger ages and accelerate at older ages as life expectancy rises; extensions to Lee-Carter explicitly model this age pattern shift. For pension systems, forecast uncertainty matters as much as point forecasts; confidence intervals translate directly into ranges for expected benefit duration and the older age population, which translates into uncertainty about future pension expenditures.

Future research is necessary to find an even better estimation method for the Lee-Carter model, possibly using machine learning or ensemble approaches and, as additional years of data accumulate, re-estimation and validation.

2. Report for statisticians

In this study we compared two methods that are used to forecast the age-specific mortality rates across countries and years. The two methods are basically two estimation approaches of the Lee-Carter model, the SVD and the Poisson MLE to handle the stochastic nature of the mortality data. Our goal here is to evaluate the performance of these methods and to find out which one performs better on the given data and whether relative performance varies by country group, sex and age range.

The Lee-Carter model (Lee & Carter, 1992) decomposes the logarithm of the mortality rate into three components plus a residual: the average log-mortality at each age across time (a_x), the sensitivity of mortality at given ages to temporal changes (b_x), the overall level of mortality that changes over time (k_t) and the residual error ($\varepsilon_{x,t}$). The formula is the following:

$$\ln(m_{x,t}) = a_x + b_x k_t + \varepsilon_{x,t}$$

The original approach uses Singular Value Decomposition (SVD) to calculate these values. However, mortality data are stochastic in nature and heteroskedastic, so an alternative is to estimate the same Lee-Carter structure by maximizing a Poisson likelihood for deaths (Brouhns et al., 2002). The expected number of deaths is $E_{x,t}m_{x,t}$, where $E_{x,t}$ is exposure to risk:

$$D_{x,t} \sim Poisson(E_{x,t}m_{x,t})$$

A primary theoretical advantage of the MLE approach over SVD is the ability to quantify parameter uncertainty. By inverting the Hessian matrix at the optimum, we can obtain standard errors and construct confidence intervals for a_x, b_x and k_t, which is less direct with the standard SVD workflow (often requiring additional assumptions or resampling such as bootstrapping). For the last step in the case of both methods, we model k_t as a stochastic time series (ARIMA) and forecast it separately for each country and sex.

We used the Human Mortality Database's *Death rates (mx,t)*, *Exposure to risk (Ex,t)* and *Number of deaths (Dx,t)* data that covered 30 countries and both sexes. Data from 1960 to 2004 was used to train our models and we evaluated them on the test data from 2005 to 2019. We excluded 2020 and above years from our test set to avoid distortions from COVID-19 mortality shocks.

Because exposures above age 90 are sparse, we aggregated ages 90+ into a single open-ended group for each country, sex, and year to reduce instability in the highest ages. We linearly interpolated missing values to get a complete age-by-year grid for estimation.

As the number of parameters is high (age groups and years) we also maximize the Poisson log-likelihood (equivalently minimize the negative log-likelihood) using BFGS method (because that uses gradients to find a local minimum of the negative log-likelihood) along with R's default Nelder-Mead method and SANN as well.

After we fitted the two versions of the Lee-Carter model on the training set, we forecasted k_t for 2005-2019 (15 years) separately for each country-sex model of the time-series holding a_x and b_x fixed at their estimated values, we generated forecasts of $m_{x,t}$ and evaluated our two models on the test data by their RMSE.

Although we used gradient-based optimizer (BFGS) for the MLE Lee-Carter model, the SVD still showed better performance on the test data. One reason behind this can be the rotation, which refers to the age pattern of mortality improvement shifting over time towards older ages in high longevity settings. In developed countries the death rates are declining in every age group, however the speed of this decline is different among young and elderly people. Most causes of infant and child deaths are treated already (by vaccinations) thus it is hard to achieve any significant improvement in this age group. On the other hand, it is much easier to expand the lifetime of the people by better medical care and with better life quality through their life so the death rates and declining in a greater speed among the elderly population. The effect of this phenomenon is that while the mortality rates are changing, the age pattern of improvement is not constant over time (Li et al., 2013). Standard Lee-Carter assumes a fixed b_x, so unmodelled rotation can degrade long-horizon projections regardless of the estimator. In our implementation, the MLE fit appeared more sensitive to these shifts and to older-age variability. For longer test data this difference might make the MLE version more accurate than the SVD (especially in the case of developed European countries). However, in a 15-year long period the SVD's constant estimation performs just as fine while the MLE forecasts were most likely more sensitive to irregular variation in the training sample.

As our main goal is to decide which method is better to predict the mortality rates we examined the question in numerous aspects. We did not want to assume that all countries and both sexes have the same pattern across the world thus we decomposed the analysis into smaller fractions. We examined mortality rates among men and women and among regions in Europe like Mediterranean or post-socialist countries.

The table below shows what percentage of the cases Poisson MLE or the SVD model was better based on RMSE. We have 60 subgroups in the dataset (30 countries each with 2 sexes) and the SVD seems to perform better in each aspect shown in Table 1. Earlier we assumed that there can be some differences between the Mediterranean and post-socialist

regions, due to their different population patterns (e.g. many more men die before they reach their retirement age in post-socialist countries or that the people generally tend to live longer in the South).

Despite these differences, SVD has lower RMSE than Poisson MLE in both regions: the SVD models have smaller RMSE in 85% of the cases, which means that in about 5 out of 6 cases, SVD was the more accurate forecaster. In Table 1, the percentages represent the share of country-sex combinations where one method produced a smaller forecast error than the other.

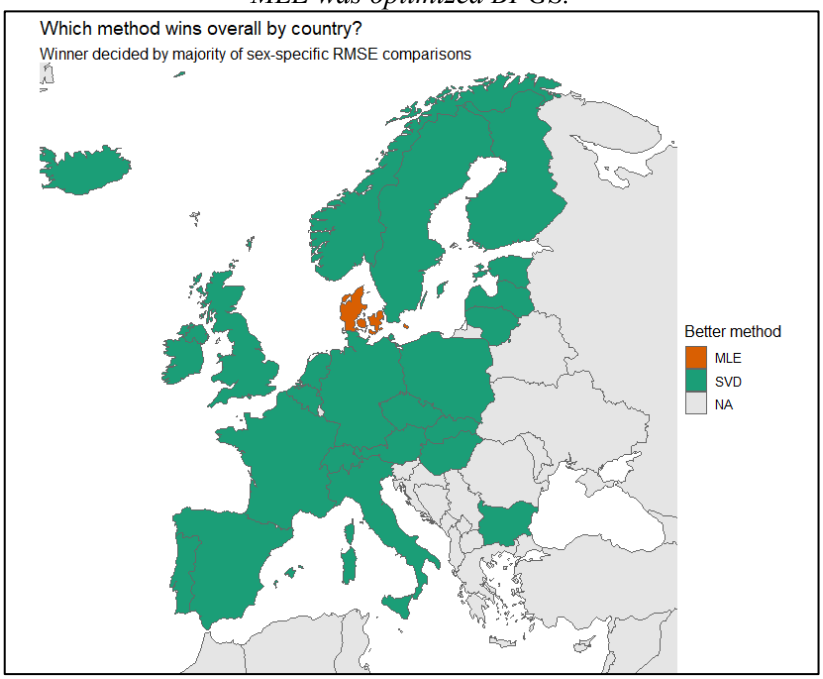
Table 1: Win Rate of Estimation Methods (% of cases where the method achieved a lower RMSE). MLE was optimized BFGS.

Type	Poisson MLE-BFGS	SVD
Overall	15%	85%
Males	13.3%	86.7%
Females	16.7%	83.3%
Mediterranean	16.7%	83.3%
Post-socialist	12.5%	87.5%

Source: Authors' calculations based on HMD data and our forecasts

The first plot shows that in vast majority of the countries, the SVD method has better performance compared to the Poisson MLE method. Denmark is the only exception where the Poisson MLE estimation of the Lee-Carter model is better in forecasting the mortality rate for both sexes. Our analysis suggests that while older populations provide the most data, they are also the most volatile due to short-term shocks (e.g. flu seasons). The MLE method tends to overreact to this noise, mistaking these temporary spikes for permanent trends. SVD acts as a stabilizer, anchoring the forecast to the smoother trends seen in younger populations.

Plot 1: Better method by country (winner determined by majority of sex-specific RMSE comparisons). MLE was optimized BFGS.



Source: Authors' own visualization

We computed our MLE model using three different optimization methods. The BFGS, Nelder–Mead, and SANN optimization methods represent three different optimization strategies for estimating the Poisson-based Lee–Carter model. BFGS relies on gradient information and therefore can converge quickly when the objective is well-scaled and initialization is good. However, it is sensitive to numerical instability and may become trapped in poor local minima if the initial values are not sufficiently close to the optimum.

The Nelder–Mead method does not require derivatives, making it usually more robust in the presence of noisy or irregular objective functions. The cost of this robustness is slower convergence and reduced efficiency in high-dimensional parameter spaces.

SANN is a stochastic, global optimization algorithm that may escape local minima, but results depend on tuning and repeated runs. This property is particularly useful when the likelihood surface is complex and highly non-convex. On the other hand, SANN is computationally expensive, produces results that may vary across runs, and is less precise in local optimization.

Overall, BFGS performs best in stable settings with good initialization, Nelder–Mead offers a reliable compromise between stability and efficiency, while SANN is better suited for robustness checks rather than as a primary estimation method.

Table 2: Comparison between SVD and Poisson MLE using three optimization methods

	Winner*		Winner*		Winner*
SVD	51	SVD	37	SVD	25
BFGS	9	Nelder-Mead	23	SANN	25**

Source: Authors' calculations

** Counts out of 60 country-sex pairs; ** Plus 10 cases resulted in a tie*

In conclusion, based on our results, we recommend the Singular Value Decomposition estimation of the Lee-Carter model to forecast mortality rates, especially in case of predicting data on 10–20-year horizons. Our training period (1960–2004) is long enough to estimate the model stably, however it was restricted to European countries. The sample consists of broadly comparable high-income countries (e.g., relatively high life expectancy and aging populations), so strong differences in performance across are not unexpected. Performance may differ in lower-income settings (e.g., parts of Asia or Africa) with different mortality and data quality. Also, over time, results may change in Europe as well, for example due to rotation, i.e., shifts in the age pattern of mortality improvement at high life expectancy. But our initial research question is answered: in our European setting (2005–2019), the SVD-based Lee-Carter estimates performed better than our Poisson MLE implementation across most country-sex groups and the region divisions considered.

Despite weaker point-forecast RMSE here, Poisson MLE remains attractive because it naturally delivers approximate confidence intervals for parameters and projected mortality, which is useful for risk analysis.

Since the publication of the original Lee-Carter model many improvements have been made in this field of research. We could expand our research by moving from a single-population model to a multi-population Lee-Carter model, which considers a common mortality pattern between multiple observed populations. Furthermore, including the logarithm of real GDP per capita as an additional factor could also create possibly more accurate forecasts.

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Statement on the use of artificial intelligence

Artificial intelligence tools were used for language editing and structural refinement of the written report, including improving clarity, grammar, and consistency of terminology. All statistical modelling, data processing, estimation, forecasting, and figure and table generation were performed by the authors in R using Human Mortality Database data. The authors reviewed and verified all technical claims, interpretations, and numerical results, and take full responsibility for the content. No confidential or restricted data were provided to the AI tool.